

Lagrangian Duality

제약이 있는 optimization 문제는 gradient가 0인 지점으로 solution 찾기 어려움

Constrained optimization problem

$$\text{minimize } f(x) \quad \text{subject to } g(x) \leq 0 \quad \& \quad h(x) = 0$$

Duality solve an optimization problem $\rightarrow$ different optimization (variable)

$$\text{Lagrangian} \quad \mathcal{L}(x, \lambda, \nu) = f(x) + \sum_{i=1}^m \underbrace{\lambda_i}_{\text{Lagrangian multipliers}} \underbrace{g_i(x)}_{\leq 0} + \sum_{j=1}^p \underbrace{\nu_j}_{\text{Lagrangian multipliers}} h_j(x)$$

$$\underbrace{D(\lambda, \nu)}_{\text{optimal value의 하한}} = \inf_x \mathcal{L}(x, \lambda, \nu) \quad \text{Convex optimization}$$

optimal value의 하한 $\rightarrow$ 최대화

Conjugate Prior posterior를 closed form으로 구하기 위해

$$p(\theta|x) \propto p(x|\theta)p(\theta)$$

likelihood의 형태가 주어질 때 prior를 어떤 분포 family에서 고르면
posterior도 prior와 같은 분포 family에 놓임

e.g., normal normal normal

- $p(y = 1|x) = \sigma(w^\top x), \sigma(a) = \frac{1}{1+e^{-a}}$
 - likelihood (Bernoulli)
 - $p(y|x, w) = p^y(1 - p)^{1-y}$ where $p = \sigma(w^\top x)$
 - $p(y|x, w) = \sigma(w^\top x)^y(1 - \sigma(w^\top x))^{1-y}$
 - MLE objective

$$p = \sigma(w^\top x) = \frac{1}{1 + e^{-w^\top x}}$$

 - maximize likelihood over data
 - equivalent to minimizing NLL (negative log-likelihood)
 - NLL = binary cross-entropy
 - $-\log p(y|x, w) = -y \log p - (1 - y) \log(1 - p)$
 - equivalent to Binary Cross Entropy (BCE)
 - gradient: $\nabla_w \mathcal{L}(w) = X^\top (p - y)$
 - training
 - no closed-form solution in general
 - optimize with gradient descent or variants (SGD, Adam)
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Convexity

any local minimum = global minimum

- determines whether a problem has a single global minimum
 - a property of functions and optimization problems
- intuition
 - a function is convex if the line segment between any two points lies above the function
 - no local minima other than the global minimum

Convex Function

- definition

- a function f is convex if for all x, y and $\lambda \in [0, 1]$:

- $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$ Jensen inequality

- geometric interpretation

$$f[E(x)] \leq E[f(x)]$$

- bowl-shaped surface
- any descent direction eventually leads to the same minimum
 - any local minimum is a global minimum
- gradient descent is guaranteed to converge (with proper step size)

- Hessian-based Characterization

- for twice-differentiable functions
 - f is convex if its Hessian is positive semi-definite
 - $\text{Hessian} \geq 0 \rightarrow \text{convex}$
- intuition
 - eigenvalues of the Hessian represent curvature
 - all eigenvalues $\geq 0 \rightarrow$ no direction curves downward

- example

- in linear regression loss

- $\text{Hessian} = 2X^T X$

- $\nabla_w \mathcal{L} = -\frac{2}{N} X^T (y - Xw)$ linear regression 이 convex 를 만족하는 이유

- always positive semi-definite $\rightarrow$ linear regression 에서의 gradient

- why convexity matters in ML

- guarantees a unique global optimum

$$\frac{1}{N} \sum_{i=1}^N (y_i - x_i^T w)^2$$

$$\nabla_w = -\frac{2}{N} X^T (y - Xw)$$

- avoids issues like local minima, saddle points, etc.

- simplifies optimization and theoretical analysis

- v.s. non-convex problems

- neural networks, matrix factorization, deep generative models, etc.
- does not mean impossible
 - but optimization guarantees are weaker

- convex $\neq$ easy

easy 보다는 guarantee $\rightarrow$ 여러 linearity 가 계산 해석력의 수준을 보임

- large-scale convex problems can still be expensive

Lipschitz Continuity

- a notion of smoothness of a function
 - controls how fast a function can change
- definition
 - a function f is Lipschitz continuous if there exists $L \geq 0$ such that
 - $\|f(x) - f(y)\| \leq L\|x - y\|$ for all x, y
- intuition
 - the function has a bounded slope
 - L is the maximum rate of change
 - prevents the function from changing too abruptly

Lipschitz Continuous Gradient (Smoothness)

gradient = 변화가 bounded
→ 빠르게 변하지 X

- a differentiable function f has an L-Lipschitz continuous gradient if
 - $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$
- interpretation
 - the gradient does not change too fast
 - the curvature of the function is bounded
- equivalently
 - the Hessian eigenvalues are bounded:
 - $\nabla^2 f(x) \preceq LI$
- why Lipschitz continuity matters in optimization
 - guarantees stability of gradient descent
 - allows choosing a safe learning rate
 - if gradient is L-Lipschitz:
 - step size $\eta \leq \frac{1}{L}$ ensures descent